AI/ML Track

Interview Preparation Series • Professional Edition

AI / ML — Index

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In This Edition

From classical machine learning to deep learning, Transformers, and building real products with LLMs — first-principles notes in the same structure as the core topics, with hand-built diagrams and magazine PDFs.

1 Files

#	Topic	Notes	Magazine PDF
1	Machine Learning Fundamentals	md	PDF
2	Classic ML Algorithms	md	PDF
3	Deep Learning & Neural Networks	md	PDF
4	Transformers & LLMs	md	PDF
5	LLM Applications & GenAI	md	PDF
6	MLOps & ML System Design	md	PDF

= the architecture behind modern AI — highest-signal for AI/ML interviews.

2 What each covers

- **01 Fundamentals** — learning paradigms, bias–variance, regularization, evaluation metrics, gradient descent.
- **02 Classic Algorithms** — regression, trees, random forests, gradient boosting (XGBoost/LightGBM), SVM, k-NN, Naive Bayes, k-Means, PCA, ensembles.
- **03 Deep Learning** — neurons, backprop, activations, optimizers, CNNs, RNNs/LSTMs, batch norm, dropout, embeddings.
- **04 Transformers & LLMs** — self-attention (Q/K/V), the Transformer, tokenization, BERT vs GPT, pre-training, RLHF/DPO, LoRA, quantization, MoE, decoding, hallucination.
- **05 LLM Applications** — prompt engineering, RAG (chunking, embeddings, vector DBs, reranking), agents (tools, function calling, ReAct), evaluation, guardrails, cost/latency.
- **06 MLOps & ML System Design** — feature stores, distributed training, serving, drift, the ML-design framework, and worked designs (recommendation, CTR, feed ranking).

3 Reading order

`01 → 02 → 03 → 04 → 05`, with `06` as the systems capstone. For AI-focused roles, prioritize **04 + 05 + 06**.

4 Each `.md` has a coherent analogy world

School (fundamentals) · a panel of doctors (algorithms) · a learning factory line (deep learning) · a newsroom of editors (Transformers) · a research assistant in a library (LLM apps) · an airport ops center (MLOps).

> Preview & edit any edition live: `cd ../.magazine && python3 serve.py` → http://localhost:8000